SDU - University of Southern Denmark

Peer Review:
Robust Visual Question Answering with a Small Vision-Language Model

Chapter 1

Summary the plan in your own words

The project formulation is a joint project combining together Computer Vision and Natural Language Processing topics, evaluating a small vision-language model (*SmolVLM-500M-Instruct*) to answer image related questions. The authors plan to establish a zero-shot baseline, fine-tune the model using LoRA, and compare accuracy on a reformulated subset of the test set, determining whether the model perceives the meaning of the question rather than depending from the question structure or word choice.

Chapter 2

Strengths

The key strengths of this project formulation are:

- **Multimodal Task**.
  The project task is relevant to both Computer Vision and Natural Language Processing learning outcomes, as the proposed objective requires text and image manipulation techniques.
- **Well-Structured Progession**.
  Starting with the base model before applying fine-tuning provides a clean overview of the model capabilities.
- **Error Analysis**.
  Incorrect predictions are divided into Visual Understanding, Language Understanding and Interactions between the two Modalities, adding strong analytical analysis that goes beyond raw accuracy numbers.

Chapter 3

Main concerns, ordered by importance